

Pump P-101 Incident Report

1. Incident Information

Incident ID: INC-P101-2026-007

Equipment Tag: P-101

Equipment Type: Centrifugal Process Pump

Incident Date: 21 July 2026

Incident Type: Abnormally High Vibration

Severity: High

Document Status: Synthetic Demonstration Document

Revision: 1.0

Note: This incident report contains synthetic information created for the Sovereign AI Workbench demonstration. It does not represent a real industrial incident.

2. Executive Summary

On 21 July 2026, the condition-monitoring system detected an abnormal increase in vibration on Pump P-101.

The recorded vibration reached **8.7 mm/s RMS**, exceeding the critical vibration limit of **7.1 mm/s RMS** defined in the synthetic P-101 Equipment Manual.

The event was investigated using sensor information, maintenance history, equipment documentation, and physical inspection findings.

The investigation identified **bearing degradation and coupling misalignment as the primary suspected contributors** to the high-vibration condition.

The affected bearing was replaced and the coupling was realigned. Following corrective maintenance, vibration decreased to **2.6 mm/s RMS** and the pump was returned to normal operating condition.

3. Equipment Information

Equipment: P-101

Equipment Type: Centrifugal Process Pump

Rated Speed: 1450 RPM

Rated Flow: 120 m³/h

Motor Power: 30 kW

Normal Vibration: Below 4.5 mm/s RMS

Warning Range: 4.5–7.1 mm/s RMS

Critical Range: Above 7.1 mm/s RMS

4. Incident Detection

The incident was initially detected through the equipment sensor-monitoring system.

Sensor Readings During the Event

Parameter	Reading	Assessment
Vibration	8.7 mm/s RMS	Critical
Temperature	71 °C	Elevated
Discharge Pressure	4.1 bar	Normal
Speed	1450 RPM	Normal

The high vibration reading was significantly above the normal operating range.

5. Initial Symptoms

The following conditions were observed:

- Significant increase in vibration
- Elevated equipment temperature
- No significant pressure deviation
- Pump speed remained approximately constant
- Abnormal mechanical condition suspected

The vibration measurement was verified before initiating further investigation.

6. Immediate Response

The following actions were taken:

1. The abnormal sensor reading was verified.
 2. The responsible operator was notified.
 3. Recent operating conditions were reviewed.
 4. Previous P-101 maintenance records were reviewed.
 5. The equipment was placed under controlled investigation.
 6. Maintenance personnel were requested to inspect the pump.
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7. Investigation

The investigation included:

Sensor Data Review

The sensor data showed a significant increase in vibration compared with previous normal readings.

Maintenance History Review

Previous maintenance records showed that P-101 had experienced an earlier increase in vibration associated with coupling alignment.

Physical Inspection

The following findings were recorded:

- Bearing wear was observed.
 - Coupling alignment was outside the acceptable condition.
 - Mechanical seal showed no major abnormality.
 - Pump mounting bolts were secure.
 - No major casing damage was identified.
 - Impeller condition was visually acceptable.
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8. Root Cause Assessment

Based on the available evidence, the primary suspected contributors were:

1. Bearing Degradation

The inspected bearing showed signs of wear that could contribute to increased vibration.

2. Coupling Misalignment

The coupling was found to be misaligned, which could contribute to abnormal vibration during operation.

The incident should not be attributed to a single cause based only on the sensor reading. The root-cause assessment was based on the combination of sensor data, maintenance history, and physical inspection findings.

9. Corrective Actions

The following actions were completed:

1. Pump was safely isolated.
2. Affected bearing was removed.
3. Replacement bearing was installed.
4. Coupling alignment was corrected.
5. Coupling fasteners were inspected and secured.
6. Bearing lubrication was completed.
7. Pump rotation was checked.

8. Pump was restarted under controlled conditions.
9. Post-maintenance sensor readings were monitored.

10. Before and After Measurements

Parameter	Before Maintenance	After Maintenance
Vibration	8.7 mm/s RMS	2.6 mm/s RMS
Temperature	71 °C	65 °C
Discharge Pressure	4.1 bar	4.2 bar
Speed	1450 RPM	1450 RPM

The reduction in vibration following corrective maintenance indicated that the maintenance actions were effective in restoring normal operating behavior.

11. Final Equipment Condition

After corrective maintenance:

Vibration: 2.6 mm/s RMS

Temperature: 65 °C

Pressure: 4.2 bar

Speed: 1450 RPM

The pump was returned to normal operating condition.

Continued monitoring was recommended to identify any recurrence of abnormal vibration.

12. Preventive Actions

The following preventive actions were recommended:

- Continue regular vibration monitoring.
 - Review vibration trends instead of relying only on individual readings.
 - Inspect coupling alignment during scheduled maintenance.
 - Monitor bearing condition.
 - Maintain appropriate lubrication.
 - Review maintenance history when new anomalies occur.
 - Investigate repeated vibration increases.
 - Maintain complete records of maintenance activities.
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13. Lessons Learned

The incident demonstrated the importance of combining multiple sources of information.

The sensor data identified the abnormal condition, but the sensor data alone did not establish the root cause.

Maintenance history provided historical context, while physical inspection provided evidence of bearing degradation and coupling misalignment.

Equipment documentation and SOP guidance can be used to determine appropriate investigation and response actions.

14. AI System Evidence Context

This synthetic incident is designed for use with the Sovereign AI Workbench.

The system can combine:

Sensor Data

Vibration reached 8.7 mm/s RMS.

Data Analytics

High-vibration anomaly detected.

Equipment Manual

Vibration above 7.1 mm/s RMS is classified as critical.

Maintenance Report

Previous P-101 vibration was associated with coupling alignment issues.

SOP

Critical vibration requires verification, investigation, and appropriate controlled response.

Incident Report

The July 2026 event involved high vibration, bearing degradation, and coupling misalignment.

These sources can be retrieved and compared by the AI system to produce an evidence-supported investigation.

15. Document Control

Incident ID: INC-P101-2026-007

Equipment: P-101

Document: Incident Report

Revision: 1.0

Purpose: Synthetic demonstration and AI/RAG testing

Record Type: Demonstration data only

End of Document